

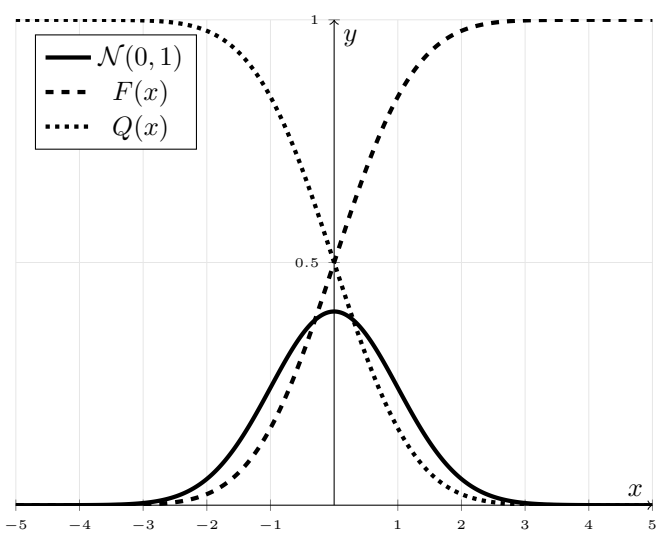
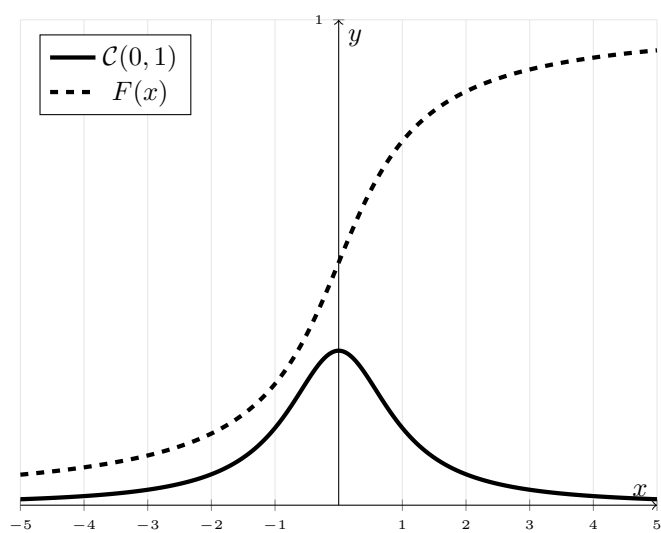
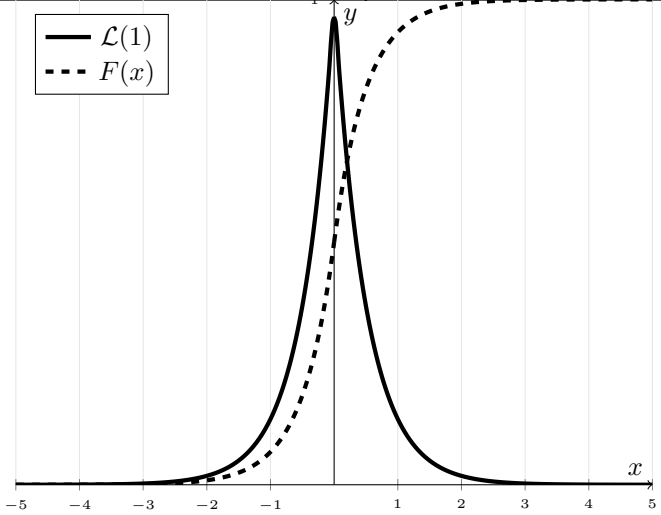
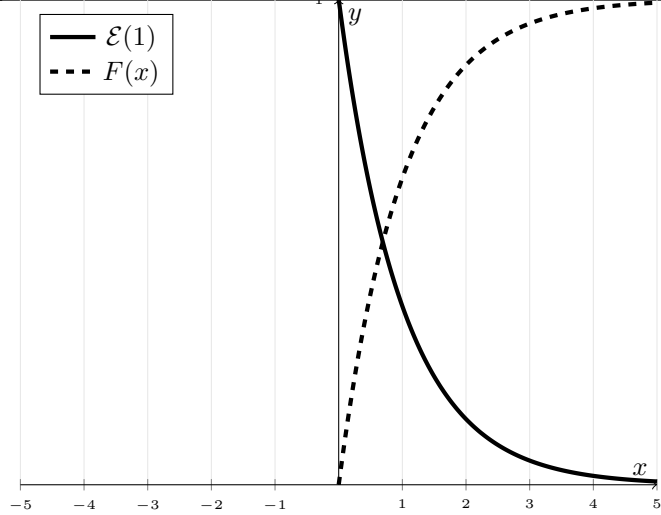
Probabilità totale in una PMF		$p_X(x) = \sum_{m=1}^{ M } \mathbb{P}(E_m)p_{X E_m}(x), \bigcup_{i=1}^m E_i = \Omega \wedge (E_i \cap E_j = \emptyset, \forall i \neq j \in \{1, \dots, m\})$			
Probabilità Condizionata		$p_X(x)p_{Y X}(y x) = p_{X,Y}(x,y) = p_Y(y)p_{X Y}(x y)$ cond fisso $\Rightarrow p_{X Y}(x y)$ è legge di prob			
Probabilità var indipendenti		$p_{Y X}(y x) = p_Y(y) \qquad p_{X Y}(x y) = p_X(x)$			
Regola della Catena		$p_{X,Y,Z}(x,y,z) = p_X(x)p_{Y X}(y x)p_{Z X,Y}(z x,y)$			
Linearità della Media		$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$			
Media Condizionata 1 var		$\mathbb{E}[X] = \sum_{m=1}^{ M } \mathbb{P}(E_m) \sum_{x \in \mathcal{X}} xp_{X E_m}(x E_m) = \sum_{m=1}^{ M } \mathbb{P}(E_m)\mathbb{E}[X E_m]$			
Media Condizionata 2 var		$\mathbb{E}[g(X,Y)] = \sum_{y \in \mathcal{Y}} p_Y(y) \sum_{x \in \mathcal{X}} g(x,y)p_{X Y}(X Y) = \sum_{y \in \mathcal{Y}} \mathbb{E}[g(X,Y) Y]p_Y(y) = \mathbb{E}[\mathbb{E}[g(X,Y) Y]]$			
Media di funzioni		$\mathbb{E}[g(x)] = \sum_{x \in \mathcal{X}} g(x)p_X(x)$			
Quantità importanti 1 var		$X_{rms}^2 = \mathbb{E}[X^2] = \sum_{x \in \mathcal{X}} x^2 p_X(x) \qquad \sigma_X^2 = \mathbb{E}[(X - \mu_X)^2] = X_{rms}^2 - \mu_X^2$			
Quantità importanti 2 var		$R_{X,Y} = \mathbb{E}[XY] \qquad COV[X,Y] = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = R_{X,Y} - \mu_X \mu_Y$ $COV(X,Y) = \rho_{X,Y} \sigma_X \sigma_Y, \rho \in [-1, 1] \qquad p_{X,Y}(x,y) = p_X(x)p_Y(y) \Rightarrow COV(X,Y) = 0$			
Combinazione lineare 2 var		$Z = aX + bY + c, \sigma_Z^2 = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2abCOV(X,Y)$			
Disuguaglianza di Chebyshev		$X > 0, \mathbb{P}\{ X - \mu_X \leq k\sigma_X\} \geq 1 - \frac{1}{k^2}$			
PMF Congiunta $\Rightarrow$ marginali		dato $p_{X,Y}(x,y), p_X(x) = \sum_{y \in \mathcal{Y}} p_{X,Y}(x,y)$			
Funzione gen Momenti $M_X(s)$		$M_X(s) = \mathbb{E}[e^{sX}] \qquad M_X(0) = 1 \qquad \frac{d^r}{ds^r} M_X(s) _{s=0} = \mathbb{E}[X^r]$ $Z = aX + bY, M_Z(s) = M_X^a(s)M_Y^b(s)$			
Proprietà prob condizionata		dati X, Y staticamente indipendenti e $Z = X + Y, p_{Z X}(z x) = p_Y(z - x)$			
Nome		PMF		$\mu \quad \sigma^2$	Proprietà
Uniforme discreta $\mathcal{X} = \{1, \dots, N\}$		$p(k) = \frac{1}{N}$		$\mu = \frac{N+1}{2} \quad \sigma^2 = \frac{N^2-1}{12}$	
Bernoulli $\mathcal{X} = \{0, 1\}$		$p(k) = \begin{cases} 1-p, & k=0 \\ p, & k=1 \end{cases}$		$\mu = p \quad \sigma^2 = p(1-p)$	
Binomiale $X \sim \mathcal{B}(n, p)$ $\mathcal{X} = \{0, 1, \dots, n\}$		$p(k) = \binom{n}{k} (1-p)^{n-k} p^k$		$\mu = np \quad \sigma^2 = np(1-p)$	
Geometrica $X \sim \mathcal{G}(p) \quad \mathcal{X} = \mathbb{N}$		$p(k) = (1-p)^{k-1} p$		$\mu = \frac{1}{p} \quad \sigma^2 = \frac{1-p}{p^2}$	
Poisson $X \sim \mathcal{P}(\lambda) \quad \mathcal{X} = \mathbb{N}_0$		$p(k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \lambda > 0$		$\mu = \lambda \quad \sigma^2 = \lambda$	$X_1 \sim P(\lambda_1), X_2 \sim P(\lambda_2)$ $Y = aX_1 + bX_2 \sim P(a\lambda_1 + b\lambda_2)$
Nome		PDF		$\mu \quad \sigma^2$	CDF
Uniforme continua $X \sim \mathcal{U}(a, b)$ $\mathcal{X} = [a, b]$		$(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b] \\ 0, & \text{else} \end{cases}$		$\mu = \frac{a+b}{2} \quad \sigma^2 = \frac{(b-a)^2}{12}$	$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x \geq b \end{cases}$
Esponenziale $X \sim \mathcal{E}(\lambda)$ $\mathcal{X} = \mathbb{R}_0^+$		$f(x) = \lambda e^{-\lambda x} u(x) \quad \lambda > 0$		$\mu = \frac{1}{\lambda} \quad \sigma^2 = \frac{1}{\lambda^2}$	$F(x) = (1 - e^{-\lambda x})u(x)$
Laplaciana $X \sim \mathcal{L}(\lambda)$ $\mathcal{X} = \mathbb{R}$		$f(x) = \frac{\lambda}{2} e^{-\lambda x } \quad \lambda > 0$		$\mu = 0 \quad \sigma^2 = \frac{2}{\lambda^2}$	$F(x) = \begin{cases} \frac{1}{2} e^{\lambda x}, & x \leq 0 \\ 1 - \frac{1}{2} e^{-\lambda x}, & x \geq 0 \end{cases}$
Cauchy $X \sim \mathcal{C}(a, b) \quad \mathcal{X} = \mathbb{R}$		$f(x) = \frac{1}{\pi b} \frac{1}{1 + (\frac{x-a}{b})^2}$		$\mu = \text{non definita}$ $\sigma^2 = \text{non definita}$ $Sym_{[-H,H]} = a$	$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x-a}{b}\right)$
Rayleigh $X \sim \mathcal{R}(\beta)$ $\mathcal{X} = \mathbb{R}_0^+$		$f(x) = \frac{x}{\beta^2} e^{-\frac{x^2}{2\beta^2}} u(x) \quad \beta > 0$		$\mu = \beta \sqrt{\frac{\pi}{2}}$ $\sigma^2 = \frac{4-\pi}{2} \beta^2$	$F(x) = 1 - e^{-\frac{x^2}{2\beta^2}}$
Nome	PDF	$\mu \quad \sigma^2$	CDF	Proprietà	
Gaussiana $X_0 \sim \mathcal{N}(0, 1)$ $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$	$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$	$\mu = \mu$ $\sigma^2 = \sigma^2$	$F_X(x) = 1 - Q(x)$ $Q(x) = \mathbb{P}(X_0 \geq x)$	$X \sim \mathcal{N}(\mu_X, \sigma_X^2) \Rightarrow X = \sigma_X X_0 + \mu_X$ $X \sim \mathcal{N}(\mu_X, \sigma_X^2) \Rightarrow \mathbb{P}(X \geq x) = Q\left(\frac{x-\mu_x}{\sigma_x}\right)$ $Q(-x) = 1 - Q(x)$ $\rho_{1,2} = 0 \Leftrightarrow f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1)f_{X_2}(x_2)$ marginali $\iff$ congiunta	
$Y = g(X), g$ continua e iniettiva (invertibile)		PDF			CDF
crescente		$f_Y(y) = \frac{f_x(g^{-1}(y))}{ g'(g^{-1}(y)) }$			$F_Y(y) = F_x(g^{-1}(y))$
decrescente		$f_Y(y) = \frac{f_x(g^{-1}(y))}{ g'(g^{-1}(y)) }$			$F_Y(y) = 1 - F_x(g^{-1}(y))$

Matrice di covarianza $\mathbf{K}_{\mathbf{x}}$: $\overline{\overline{\mathbf{K}_{\mathbf{x}}}} := E \left[\left(\overline{X} - \overline{\mu}_x \right) \left(\overline{X} - \overline{\mu}_x \right)^T \right] = \begin{pmatrix} \sigma_1^2 & \sigma_1 \sigma_2 \rho_{1,2} \\ \sigma_1 \sigma_2 \rho_{1,2} & \sigma_2^2 \end{pmatrix}$

Proprietà della Matrice di covarianza $\mathbf{K}_{\mathbf{x}}$
 $|\mathbf{K}_{\mathbf{x}}| = \sigma_1^2 \sigma_2^2 (1 - \rho_{1,2}^2) \geq 0$
 $\rho_{1,2} \neq \pm 1 \implies \mathbf{K}_{\mathbf{x}}^{-1} = \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho_{1,2}^2)} \begin{pmatrix} \sigma_2^2 & -\sigma_1 \sigma_2 \rho_{1,2} \\ -\sigma_1 \sigma_2 \rho_{1,2} & \sigma_1^2 \end{pmatrix}$
 $z = [z_1, z_2]^T \in \mathbb{R}^{2 \times 1} \implies \overline{\overline{z}}^T \overline{\overline{K_x^{-1}}} \overline{\overline{z}} \geq 0$
 $f_X(\mathbf{x}) = \frac{1}{2\pi |\mathbf{K}_X|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_X)^T \mathbf{K}_X^{-1} (\mathbf{x} - \boldsymbol{\mu}_X) \right] = f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1 - \rho_{1,2}^2)}} \exp \left[-\frac{\sigma_2^2 (x_1 - \mu_1)^2 + \sigma_1^2 (x_2 - \mu_2)^2 - 2 \rho_{1,2} (x_1 - \mu_1)(x_2 - \mu_2)}{2 \sigma_1^2 \sigma_2^2 (1 - \rho_{1,2}^2)} \right]$

Derivata del Prodotto	$\frac{d}{dx} f(x) \cdot g(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
Derivata del Rapporto	$\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)^2]}$
Regola della catena	$\frac{d}{dx} f[g(x)] = f'[g(x)] \cdot g'(x)$
Integrazione generale	$\int f\left[g\left(x\right)\right] \cdot g'\left(x\right) dx = F\left[g\left(x\right)\right] + c$
Integrazione per parti	$\int f\left(x\right) \cdot g\left(x\right) dx = F\left(x\right) g\left(x\right) - \int F\left(x\right) \cdot g'\left(x\right) dx$
Funzione Gamma Γ	$\int_0^{+\infty} t^{x-1} e^{-at} dt = a^{-x} \Gamma(x) \qquad \Gamma(x) := \int_0^{+\infty} t^{x-1} e^{-t} dt \qquad \Gamma(n) = (n-1)!, n \in \mathbb{N}$
Sommatoria primi n numeri	$\sum_{i=1}^n i = \frac{n(n+1)}{2} \qquad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$
Serie geometrica	$\sum_{i \in \mathbb{N}_0} p^i = \frac{1}{1-p}, p < 1 \qquad \sum_{i=0}^{n-1} p^i = \frac{1-p^n}{1-p}$ $0 \leq p < 1, \sum_{n=1}^{+\infty} np^{n-1} = \frac{1}{(1-p)^2} \sum_{n=1}^{\infty} n^2 q^{n-1} = \frac{1}{(1-q)^2} + \frac{2q}{(1-q)^3}$

Funzione	Derivata	Integrale
x^n	nx^{n-1}	$\frac{x^{n+1}}{n+1} + c, n \neq -1$
$\frac{1}{x}$	$-x^{-2}$	$\log(x) + c$
e^x	e^x	$e^x + c$
a^x	$a^x \log(a)$	$a^x \log_a(e) + c$
$\cos(x)$	$-\sin(x)$	$\sin(x) + c$
$\sin(x)$	$\cos(x)$	$-\cos(x) + c$
$\frac{1}{1+x^2}$	$-\frac{2x}{(1-x^2)^2}$	$\arctan(x) + c$



Definizione informazione	$i(A) = \log\left(\frac{1}{\mathbb{P}(A)}\right) = -\log(\mathbb{P}(A))$
Definizione entropia	$H(X) = \mathbb{E}[i(X)] = \sum_{x \in \mathcal{X}} p_X(x) \log\left(\frac{1}{p_X(x)}\right)$
Entropia Bernoulli	$H(p) = H_2(p) = H(p, 1-p) = p \log\left(\frac{1}{p}\right) + (1-p) \log\left(\frac{1}{1-p}\right) = -p \log(p) - (1-p) \log(1-p) \quad \frac{d}{dp} H(p) = \log\left(\frac{1-p}{p}\right)$
Entropia uniforme discreta	$H(X) = \log(\mathcal{X}), \mathcal{X}$ uniforme discreta
Disuguaglianza di Jensen	$g(x)$ differenziabile nei punti dell'alfabeto $\mathcal{X}$ e convessa $\Rightarrow \mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$, $g(x)$ differenziabile nei punti dell'alfabeto $\mathcal{X}$ e concava $\Rightarrow \mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$
Limite superiore Entropia	$H(X) \leq \log(\mathcal{X})$
Divergenza KL (Kullback-Leibler)	$p(x), q(x)$ pmf su $\mathcal{X}$, $D(p q) = \sum_{x \in \mathcal{X}} p(x) \log\left(\frac{p(x)}{q(x)}\right) = \mathbb{E}_{p(x)} \left[\log\left(\frac{p(x)}{q(x)}\right) \right]$, $D(p q) \geq 0 \quad \forall p, q, D(p q) = 0 \iff p = q$
Disuguaglianza di Kraft	$\sum_{x \in \mathcal{X}} 2^{-l(x)} \leq 1 \iff$ il codice è a prefisso, $l(x)$ lunghezza codice associato a x
Lunghezza di Shannon	$l(x) = \left\lceil \log\left(\frac{1}{p_X(x_i)}\right) \right\rceil$, esiste almeno un codice istantaneo con le lunghezze di Shannon. Inoltre $H(X) \leq L = \mathbb{E}_X[l(X)]$
Entropia coppie variabili	$H(X, Y) = H(X) + H(Y X) = H(Y) + H(X Y), 0 \leq H(X Y) \leq H(X), \forall Y$, con $H(X Y) = H(X) \iff X, Y$ indipendenti e $H(X Y) = 0 \iff X = g(Y)$
Probabilità totale Entropia	$H(X Y) = \sum_{y \in \mathcal{Y}} \mathbb{P}(Y = y) H(X Y = y)$
Mutua informazione	$I(X; Y) = H(X) - H(X Y) = H(Y) - H(Y X) = \mathbb{E} \left[\log \frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)} \right] =$ $D(p_{X,Y}(x,y) p_X(x)p_Y(y)), \quad I(X; Y) = 0 \iff p_{X,Y}(x,y) = p_X(x)p_Y(y)$
Entropia vettore aleatorio	$H(X^n) = H(X_1, \dots, X_n) = H(X_1) + H(X_2 X_1) + \dots + H(X_n X_{n-1}, X_{n-2}, \dots, X_1)$
Tasso entropico	$H_\infty(X) = \lim_{n \rightarrow \infty} \frac{H(\mathbf{X}^n)}{n} = \lim_{n \rightarrow \infty} H(X_n X_{n-1}, X_{n-2}, \dots, X_1)$
Sorgenti Markoviane	$p_{X(n) X(n-1), \dots, X(1)}(x_n x_{n-1}, \dots, x_1) = p_{X(n) X(n-1)}(x_n x_{n-1})$
Matrice di transizione	dato $\mathcal{X} = \{1, \dots, m\}, \overline{P} \in \mathcal{M}_{m \times m}$ tale che $\forall i, j \in \mathcal{X} : p_{i \rightarrow j}$ è la probabilità di andare dallo stato i allo stato j
Vettore di probabilità degli stati a tempo n	$\bar{x}(n_o + n) = (p_{X(n_o+n)}(1), \dots, p_{X(n_o+n)}(m)) = (\overline{P^T})^n \bar{x}(n_o)$
Condizione di stazionarietà	$\bar{x}$ è distribuzione stazionaria (indipendente da n) $\iff \overline{P^T} \bar{x} = \bar{x}$
Distribuzione stazionaria grafo NON orientato	$\bar{x} = (\frac{w_1}{2w}, \dots, \frac{w_m}{2w}), w_i =$ somma pesi archi uscenti da S_i e w somma pesi tutti gli archi.
Tasso entropico Markov	$H_\infty(X) = \sum_{i=1}^m \bar{x}_i \sum_{j=1}^m p_{i \rightarrow j} \log\left(\frac{1}{p_{i \rightarrow j}}\right) = \sum_{i=1}^m \bar{x}_i H(S_i)$, dove $H(S_i)$ è l'incertezza di uscire dallo stato i

Bias/unbias	$\hat{\Theta}$ unbiased/corretto $\iff \mathbb{E}[\hat{\Theta}(X^n) \Theta = \theta] = \theta$
unbias/corretto asintotico	$\hat{\Theta}$ unbiased/corretto asintoticamente $\iff \lim_{n \rightarrow \infty} (\mathbb{E}[\hat{\Theta}(X^n) \Theta = \theta]) = \theta$
Consistenza in probabilità	$\lim_{n \rightarrow \infty} \mathbb{P} \left\{ \left \hat{\Theta}(X^n) - \Theta \right > \epsilon \right\} = 0, \forall \epsilon > 0$
Consistenza in media quadratica	$\hat{\Theta}$ consistente in media quadratica $\iff \lim_{n \rightarrow \infty} (\bar{e}^2) = 0, \bar{e}^2 = \mathbb{E}[(\hat{\Theta}(X^n) - \Theta)^2]$ se $\hat{\Theta}$ è unbiased $\bar{e}^2 = Var(\hat{\Theta}(X^n))$
Informazione di Fisher	$I_n = \mathbb{E} \left[\left(\frac{d \log(f_{X^n}(x^n; \theta))}{d\theta} \right)^2 \right] = -\mathbb{E} \left[\frac{d^2 \log(f_{X^n}(x^n; \theta))}{d^2 \theta} \right]$
Limite CR	$Var[\hat{\Theta}(X^n)] \geq \frac{1}{I_n(\theta)}$ con $\hat{\Theta}(X^n)$ efficiente se vale '='

NOTA: Se in uno stimatore vale che $C(\cdot) \geq 0$ e convessa ($\cup$), $f_{\Theta|X^n}(\theta|x^n)$ è simmetrica rispetto a $\mathbb{E}[\Theta|X^n = x^n]$ e $f_{X^n}(x^n; \theta)$ è unimodale, Allora tutti gli stimatori che ottimizzano $C(\cdot)$ coincidono con $\mathbb{E}[\Theta|X^n]$ e, quindi equivalenti a $\hat{\Theta}_{mmse}(X^n) = \hat{\Theta}_{map}(X^n)$.
NOTA: APPLICARE IL LOGARITMO NATURALE NON CAMBIA IL VERSO DELLA DISEQUAZIONE, FORTEMENTE CONSIGLIATI IN TUTTI GLI STIMATORI.

Tipo inf	Cose note	Regola decisione/Stimatore		
Decisione Bayesiana	$\{H_i\}_{i=1}^M$ disgiunte $\forall i \in \{1, \dots, M\} : \mathbb{P}\{H_i\}$ $\forall i \in \{1, \dots, M\} : \mathbb{P}\{X^n = x^n H_i\}$ $\overline{C} \in \mathcal{M}_{m \times m} : c_{i,j}$ costo se decido i ma è vera l'ipotesi H_j	$\mathcal{R} = \sum_{i=1}^M \sum_{j=1}^M C_{i,j} \mathbb{P}\{D(X^n) = i, H = H_j\}$ $D(x^n) = i \iff \mathbb{P}\{X^n = x^n, H_i\} > \mathbb{P}\{X^n = x^n, H_j\} \forall j \neq i$ $L(x^n) = \frac{\mathbb{P}\{X^n=x^n H_1\}}{\mathbb{P}\{X^n=x^n H_2\}} \underset{H_2}{\overset{H_1}{\geq}} \frac{\mathbb{P}\{H_2\}}{\mathbb{P}\{H_1\}}$ con 2 ipotesi		
Neyman-Pearson	H_0 stato normale, $H_1 = \overline{H_0}$ stato alterato $\forall i \in \{0, 1\} : \mathbb{P}\{X^n = x^n H_i\}$ Errore di tipo I: $P\{D(X^n) = 1 H_0\}$ Potenza del test: $P\{D(X^n) = 1 H_1\}$	$L(x^n) = \frac{\mathbb{P}\{X^n=x^n H_1\}}{\mathbb{P}\{X^n=x^n H_0\}} \underset{H_0}{\geq} \eta$ $\alpha = \mathbb{P}\{L(x^n) > \eta H_0\}$ $1 - \beta = \mathbb{P}\{L(x^n) > \eta H_1\}$		
Stima Bayesiana parametro continuo	$f_{\Theta X^n}(\theta x^n) = \frac{f_{\Theta}(\theta)p_{X^n \Theta}(x^n \theta)}{p_{X^n}(x^n)}$ $p_{X^n}(x^n) = \int f_{\Theta}(\theta)p_{X^n \Theta}(x^n \theta)d\theta$ $\mathcal{R} = E\left[C\left(\hat{\Theta}(X^n) - \Theta\right)\right] =$ $E_{x^n}\left[E\left[C\left(\hat{\Theta}(X^n) - \Theta\right) \middle X^n\right]\right]$	$\hat{\theta}_{opt}(x^n) = \arg \min \int C(\hat{\theta}(x^n) - \theta)f_{\Theta X^n}(\theta x^n)d\theta$ $C(x) = x^2 \Rightarrow \hat{\theta}_{mmse}(x^n) = \int \theta f_{\Theta X^n}(\theta x^n)d\theta = \mathbb{E}[\Theta X^n = x^n]$ $C(x) = \Pi(\frac{x}{\epsilon}) = \begin{cases} 1, & x < \frac{\epsilon}{2} \\ 0, & else \end{cases} \Rightarrow \hat{\theta}_{map}(x^n) = \arg \max f_{\Theta X^n}(\theta x^n)$		
Non Bayes	$f_{X^n}(x^n; \theta)$	$\hat{\theta}_{ML}(x^n) = \arg \max \log(f_{X^n}(x^n; \theta))$		
Nome		Entropia H	Stimatori/Likelihood dato X^n con n realizzazioni (x_i) iid	Inform. Fisher I_n
Uniforme discreta $\mathcal{X} = \{1, \dots, m\}$		$H = \log(\mathcal{X}) = \log(m)$	Stimare il parametro $m \implies$ $\hat{m} = \max\{x_1, \dots, x_n\}$	
Uniforme continua $X \sim \mathcal{U}(a, b)$ $\mathcal{X} = [a, b]$		$H = \log(b - a)$	Stimare il parametro a o $b \implies$ $\hat{a} = \min\{x_1, \dots, x_n\}$ $\hat{b} = \max\{x_1, \dots, x_n\}$	
Bernoulli $\mathcal{X} = \{0, 1\}$		$H = H(p)$	Stimare il parametro $p \implies$ $\Lambda(p) = \log(p)w(n) + \log(1-p)(n-w(n))$ $\frac{d}{dp}\Lambda(p) = \frac{w(n)}{p} - \frac{n-w(n)}{1-p} = 0$	$I = \frac{1}{p(1-p)}$ $I_n = \frac{n}{p(1-p)}$
Binomiale $X \sim \mathcal{B}(m, p)$ $\mathcal{X} = \{0, 1, \dots, m\}$			Stimare il parametro $p \implies \Lambda(p) =$ $\sum_{i=1}^n \log \binom{m}{x_i} + \log(p) \sum_{i=1}^n x_i + \log(1-p) \sum_{i=1}^n (m - x_i)$ $\frac{d}{dp}\Lambda(p) = \frac{\sum_{i=1}^n x_i}{p} - \frac{\sum_{i=1}^n (m - x_i)}{1-p} = 0$	$I = \frac{m}{p(1-p)}$ $I_n = \frac{nm}{p(1-p)}$
Geometrica $X \sim \mathcal{G}(p) \quad \mathcal{X} = \mathbb{N}$		$H = -\log(p) - \frac{1-p}{p} \log(1-p)$	Stimare il parametro $p \implies$ $\Lambda(p) = n \log(p) + \log(1-p) \sum_{i=1}^n (x_i - 1)$ $\frac{d}{dp}\Lambda(p) = \frac{n}{p} - \frac{\sum_{i=1}^n (x_i - 1)}{1-p} = 0$	$I = \frac{1}{p^2(1-p)}$ $I_n = \frac{n}{p^2(1-p)}$
Poisson $X \sim \mathcal{P}(\lambda)$ $\mathcal{X} = \mathbb{N}_0$			Stimare il parametro $\lambda \implies$ $\Lambda(\lambda) = \log(\lambda) \sum_{i=1}^n x_i - n\lambda - \sum_{i=1}^n \log(x_i!)$ $\frac{d}{d\lambda}\Lambda(\lambda) = \frac{\sum_{i=1}^n x_i}{\lambda} - n = 0$	$I = \frac{1}{\lambda}$ $I_n = \frac{n}{\lambda}$
Esponenziale $X \sim \mathcal{E}(\lambda) \quad \mathcal{X} = \mathbb{R}_0^+$		$H = 1 - \log(\lambda)$	Stimare il parametro $\lambda \implies$ $\Lambda(\lambda) = n \log(\lambda) - \lambda \sum_{i=1}^n x_i$ $\frac{d}{d\lambda}\Lambda(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$	$I = \frac{1}{\lambda^2}$ $I_n = \frac{n}{\lambda^2}$
Laplaciana $X \sim \mathcal{L}(\lambda) \quad \mathcal{X} = \mathbb{R}$		$H = 1 + \log\left(\frac{2}{\lambda}\right)$	Stimare il parametro $\lambda \implies$ $\Lambda(\lambda) = n \log\left(\frac{\lambda}{2}\right) - \lambda \sum_{i=1}^n x_i $	$I = \frac{1}{\lambda^2}$ $I_n = \frac{n}{\lambda^2}$
Cauchy $X \sim \mathcal{C}(a, b)$ $\mathcal{X} = \mathbb{R}$		$H = \log(4\pi b)$	Stimare i parametri a e $b \implies$ $\Lambda(a; b) = -n \log(b\pi) - \sum_{i=1}^n \log\left(1 + \left(\frac{x_i - a}{b}\right)^2\right)$ NO!	$I_n = \frac{1}{2b^2}$
Rayleigh $X \sim \mathcal{R}(\beta)$ $\mathcal{X} = \mathbb{R}_0^+$		$H = 1 + \log\left(\frac{\beta}{\sqrt{2}}\right) + \frac{\gamma}{2}$ $\gamma \approx 0.5772 \dots$	Stimare il parametro $\beta \implies$ $\Lambda(\beta) = -2n \log(\beta) + \sum_{i=1}^n \log(x_i) - \frac{1}{2\beta^2} \sum_{i=1}^n x_i^2$ $\frac{d}{d\beta}\Lambda(\beta) = -\frac{2n}{\beta} + \frac{1}{\beta^3} \sum_{i=1}^n x_i^2 = 0$	$I = \frac{4}{\beta^2}$ $I_n = \frac{4n}{\beta^2}$
Gaussiana $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$		$H = \frac{1}{2} (1 + \log(2\sigma_X^2 \pi))$	Stimare i parametri μ_X e $\sigma_X^2 \implies$ $\Lambda(\mu_X; \sigma_X^2) = -\frac{n}{2} \log(2\pi\sigma_X^2) - \frac{1}{2\sigma_X^2} \sum_{i=1}^n (x_i - \mu_X)^2$ $\frac{d}{d\mu_X}\Lambda(\mu_X; \sigma_X^2) = -\frac{1}{\sigma_X^2} \sum_{i=1}^n \mu_X - x_i$ $\frac{d}{d\sigma_X^2}\Lambda(\mu_X; \sigma_X^2) = -\frac{n}{2\sigma_X^2} + \frac{1}{2\sigma_X^4} \sum_{i=1}^n (x_i - \mu_X)^2$	$I^{\mu_X} = \frac{1}{\sigma_X^2}$ $I_n^{\mu_X} = \frac{n}{\sigma_X^2}$ $I^{\sigma_X^2} = \frac{1}{2\sigma_X^4}$ $I_n^{\sigma_X^2} = \frac{n}{2\sigma_X^4}$